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Offline Wi-Fi Port Communication between Raspberry Pi and Laptop

PROBLEM STATEMENT:

To create a local communication channel between a Raspberry Pi and a laptop using a WiFi module or hotspot without internet access. The communication will use TCP socket programming and a specific port number so both devices can send and receive messages.

REQUIRED HARDWARE:

- Raspberry Pi (any model with wifi)
- Laptop
- Wi-fi module / mobile hotspot / ESP32 access point
- MicroSD card

REQUIRED SOFTWARE:

- Arduino IDE
- Python3.10(64-bit)
- Raspberry pi OS
- Python Socket Library

- Terminal

WI-FI CONNECTION FOR ESP32:

```
#include <WiFi.h>

const char* ssid = "Vivo Y28";
const char* password = "_____";

void setup() {
  Serial.begin(115200);

  Serial.println("Connecting to WiFi...");

  WiFi.begin(ssid, password);

  while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
    Serial.println("Connecting...");
  }

  Serial.println("WiFi Connected!");
  Serial.print("IP Address: ");
  Serial.println(WiFi.localIP());
}

void loop() {
  // Nothing needed here
}
```

CREATE A SERVER ON RASPBERRY PI:

```
import socket

host = '10.235.28.211'
port = 5000

server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind((host, port))
server.listen(1)

print('Waiting for connection...')

conn, addr = server.accept()
print('Connected by', addr)

while True:
    data = conn.recv(1024)
    if not data:
        break
    print('Received:', data.decode())
```

Run the server:

```
python3 server.py
```

CREATE CLIENT ON LAPTOP:

```
import socket

host = '10.235.28.211'  # Raspberry Pi IP
port = 5000

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect((host, port))
```

```
message = 'Hello from laptop'
client.send(message.encode())
print('Message sent')
```

Run the Client

python3 client.py

```
    server.bind((host,port))
    ~~~~~~~~~~~^^^^^^^^^^^^^
OSError: [Errno 99] Cannot assign requested address
team7@team7:~ $ nano rasp.py
team7@team7:~ $ python3 rasp.py
Waiting for connection...
^CTraceback (most recent call last):
  File "/home/team7/rasp.py", line 12, in <module>
    conn,addr = server.accept()
                ~~~~~~~~~~~~~^^
  File "/usr/lib/python3.13/socket.py", line 295, in accept
    fd, addr = self._accept()
               ~~~~~~~~~~~~^^
KeyboardInterrupt

team7@team7:~ $ hostname -I
10.235.28.211 2409:40f4:3013:ff5d:88bf:3e7f:a83d:d2be 2409:40f4:301b:abd7:3628:37b3:9248:fd2d
team7@team7:~ $ python3 server.py
Traceback (most recent call last):
  File "/home/team7/server.py", line 7, in <module>
    client.connect((host, port))
    ~~~~~~~~~~~~~~^^^^^^^^^^^^^^
ConnectionRefusedError: [Errno 111] Connection refused
team7@team7:~ $ python3 rasp.py
Waiting for connection...
Connected by ('10.235.28.30', 53104)
Received: Hello from laptop
```

FOR EITHER SIDE COMMUNICATION

(Raspberry ←→ Laptop):

SERVER CODE:

```
import socket

host = '10.235.28.211'   # Raspberry pi IP
port = 5000

server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind((host, port))
server.listen(1)

print("Waiting for connection...")
```

```
conn, addr = server.accept()
print("Connected by", addr)

while True:
    # Receive message
    data = conn.recv(1024)
    if not data:
        break
    print("Laptop:", data.decode())

    # Send reply
    message = input("Pi: ")
    conn.send(message.encode())

conn.close()
```

CLIENT CODE:

```
import socket


host = '10.235.28.211'

port = 5000


client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

client.connect((host, port))


print("Connected to Raspberry Pi")


while True:

    # Send message

    message = input("Laptop: ")

    client.send(message.encode())


    # Receive reply
```

```
data = client.recv(1024)
print("Pi:", data.decode())
```

OUTPUT:

```
team7@team7:~ $ python3 rasp.py
Waiting for connection...
Connected by ('10.235.28.30', 53104)
Received: Hello from laptop
team7@team7:~ $ nano rasp.py
team7@team7:~ $ python3 server.py
Traceback (most recent call last):
  File "/home/team7/server.py", line 7, in <module>
    client.connect((host, port))
    ~~~~~~~~~~~~~~^^^^^^^^^^^^^^
ConnectionRefusedError: [Errno 111] Connection refused
team7@team7:~ $ python3 rasp.py
Waiting for connection...
Connected by ('10.235.28.30', 63663)
Laptop: hiii
Pi: hello
Laptop: ena [pandra
Pi: just hacking yeah!!
Laptop: ohhh
Pi: fine
```

OVERALL FLOW:

Setup the devices (lap & rasp) with same wi-fi network

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Note the IP address of both devices
For rasp: hostname -I

For lap: run the wi-fi code note the

IP from serial monitor

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Make raspberry pi as the Server

⇩

Make laptop as the Client

⇩

Run the code

Rasp: python3 server.py

Python terminal: python client.py

⇩

Communication begins between two devices without separate network access(thorough local network)

RESULT:

Thus the ESP32 successfully created a wi-fi Access point through which laptop and raspberry pi connected and communicated each other. Allowing real-time dual side communication between devices without internet access has been done through the specified port (port:5000).